

# Our workflow service, as Camunda would want to see it.

## THE ASK, IN ONE SENTENCE

**Assess the effort and risk of moving **ms-bpm** from Camunda 7.23 Community Edition to **Camunda 7 Enterprise** or to **Camunda 8 Enterprise**, so that we can choose between them.**

## WHY THIS DECK EXISTS

**We measured our own system before asking. Every element in our models is counted, every engine call in our code is listed, and the production numbers are real. The five markdown files that come with this deck hold the detail. This deck is the first meeting.**

# One Spring Boot service, one embedded engine, one database, every retail loan we write

A consumer-finance lender in Indonesia. `ms-bpm` takes a loan application from lead to go-live: survey, credit bureau, scoring, underwriting, approval, signing. It is the system of record for applications in flight.

|                                                                         |                                                                                        |                                                       |                                                                          |
|-------------------------------------------------------------------------|----------------------------------------------------------------------------------------|-------------------------------------------------------|--------------------------------------------------------------------------|
| <div>135<sub>k</sub></div> <div>LOAN APPLICATIONS STARTED A MONTH</div> | <div>541<sub>k</sub></div> <div>APPLICATIONS OPEN RIGHT NOW, TAIL UP TO ONE YEAR</div> | <div>53</div> <div>BPMN MODELS AND 3 DMN TABLES</div> | <div>272</div> <div>JAVA DELEGATE CLASSES BEHIND 483 SERVICE TASKS</div> |
|-------------------------------------------------------------------------|----------------------------------------------------------------------------------------|-------------------------------------------------------|--------------------------------------------------------------------------|

TWO GENERATIONS OF MODEL, BOTH IN SCOPE

**Legacy, per product:** five large flat processes (NDF4W, NDF2W, repeat-order, Sharia, unsecured) carry about 94% of volume. **Unified:** one spine that calls 32 child processes, one per domain step, where new products land. Product routing is a map in configuration, not a BPMN decision.

WHAT SITS AROUND THE ENGINE

About thirty microservices called from delegates. Three in-house operator consoles (764,000 lines of React) that **never see a Camunda task id**. Cockpit, used by operations to retry failed jobs. Keycloak for identity. A second, separate Camunda 7.17 in another squad, out of scope here but relevant to any licence conversation.

# The stack, and the two dates that brought us to you

| COMPONENT               | VERSION                       | NOTE                                                                                                    |
|-------------------------|-------------------------------|---------------------------------------------------------------------------------------------------------|
| Camunda Platform 7      | 7.23.0 Community Edition      | engine, webapp and REST starters embedded; DMN engine; Spin declared but unused; JUEL 2.2.7             |
| Spring Boot / Framework | 3.5.16 / 6.2.19               | Spring Cloud 2025.0, 113 Feign clients to other services                                                |
| Java                    | 17                            | distroless image, container-aware JVM at 80% of the memory limit                                        |
| Database                | PostgreSQL 16.14 on Cloud SQL | engine tables and 271 application tables share one schema and one transaction                           |
| Platform                | GKE, Helm, ArgoCD             | 2 replicas conventional (4-8 CPU, 8-16 GiB each), 4 replicas Sharia; Istio and Cloud SQL proxy sidecars |
| Identity                | Keycloak OIDC                 | custom read-only identity provider plugin; engine user tables empty                                     |
| Observability           | Datadog APM                   | HTTP, JPA and outbound calls traced; no engine-level metrics exported                                   |

*Camunda 7 Community Edition ended in October 2025. Spring Boot 3.5 open-source support ended in June 2026. We run the last public build of each. That is the whole reason for this package.*

**TWO PRODUCTION DEPLOYMENTS, SAME CODE**

**Conventional** and **Sharia**, each with its own database and promotion pipeline. Sharia is small (86 applications in 90 days, 2.9 GB) but regulated separately. Any licence, tenancy or cluster answer has to cover both.

# What is in the 53 models, counted element by element

|                                                                                              |                                                                                                  |                                                                                            |                                                                                                  |
|----------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| <div>483</div> <div>SERVICE TASKS, EVERY ONE A<br/>DELEGATEEXPRESSION TO A SPRING BEAN</div> | <div>96</div> <div>USER TASKS IN 26 FILES; NO FORMS, NO<br/>CANDIDATE GROUPS, NO LISTENERS</div> | <div>56</div> <div>CALL ACTIVITIES, ALL LATEST BINDING, ALL<br/>VARIABLES IN AND OUT</div> | <div>513</div> <div>EXCLUSIVE GATEWAYS; 14 PARALLEL; 4<br/>INCLUSIVE; 2,214 SEQUENCE FLOWS</div> |
|----------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|

THREE PATTERNS CARRY THE MODELS

**Error end event to error boundary event** on an enclosing sub-process: 104 ends, 106 boundaries, 114 error codes, thrown from 88 BpmnError sites in Java. **Escalation end to escalation boundary** for non-error outcomes such as “send back to surveyor”: 108 ends, 82 boundaries, 97 codes. **Link events** to keep big flat diagrams readable: 146 throw, 48 catch. 154 embedded sub-processes exist mostly as scopes for those boundaries.

WHAT IS NOT THERE

No script tasks, no send or receive tasks, no message or signal events, no multi-instance, no compensation, no execution or task listeners, no external-task topics, no connectors, no field injection, no tenants, no form keys. Timers are 12 fixed durations from 5 seconds to an hour. **The surface is simpler than the file count suggests.**

# All JUEL, no FEEL; user tasks are wait states, not forms

## 659

JUEL CONDITIONS ON SEQUENCE FLOWS; 109 CALL A JAVA METHOD

## 78

OF THEM READ SPRING CONFIGURATION WITH `ENVIRONMENT.GETPROPERTY(...)` TO SWITCH FLOW ON A FEATURE FLAG

## 239

INPUT AND OUTPUT PARAMETER MAPPINGS, MOSTLY STRING LITERALS

## 186

`FAILEDJOBRETRYTIMECYCLE` DECLARATIONS, 31 DISTINCT BACK-OFF PATTERNS; 260 `ASYNCATER`

### HOW A HUMAN TASK ACTUALLY WORKS HERE

Our console shows the work and calls a domain endpoint on `ms-bpm`. The service finds the task by process instance and task key, then calls `TaskService.complete`. 48 complete sites, 24 claim sites. Assignment comes from a variable (`#{surveyAssignee}`, 21 tasks). **Tasklist and Camunda Forms are not used.** We need the wait state and an API, not a UI.

### THE EXPRESSION PATTERN WITH NO OBVIOUS CAMUNDA 8 HOME

Reading a *Spring bean* inside a gateway condition works only because the engine lives in our JVM. 78 conditions do it. Turning every flag into a process variable on every instance is the naive fix. We would like your recommended pattern before we count effort.

# Thin at the API, deep in one place: the shared transaction

0.4%

OF 499,000 JAVA LINES TOUCH AN ENGINE API

1,917

SETVARIABLE SITES; 995 GETVARIABLE. THE DELEGATES ARE VARIABLE-CENTRIC

0

HISTORY QUERIES, EXTERNAL TASKS, MESSAGE CORRELATION, BUSINESS KEYS

9

FILES IMPORT ENGINE INTERNALS: ONE RETRY ASPECT, ONE IDENTITY PLUGIN, ONE UTILITY

| ENGINE SERVICE                                       | CALLS | WHAT FOR                                                                        |
|------------------------------------------------------|-------|---------------------------------------------------------------------------------|
| TaskService                                          | 96    | complete 48, claim 24, query 23                                                 |
| RuntimeService                                       | 61    | start 14, query 14, variables 21, <b>modification 5</b> , delete 4, incidents 3 |
| ManagementService                                    | 1     | set job retries from our own API                                                |
| IdentityService, RepositoryService                   | 5     | Cockpit SS0; read element names from the model                                  |
| HistoryService, DecisionService, ExternalTaskService | 0     | –                                                                               |

THE ONE DEEP DEPENDENCY

A delegate updates our JPA entities and the engine updates `ACT_RU_*` in the same **PostgreSQL transaction**. If the delegate throws, both roll back together. 260 `asyncAfter` boundaries were placed with that in mind. In a job-worker model those become two operations, and every one of 483 tasks needs an idempotency answer. We think most of it can live in one base worker class. We want your view.

# Long-running instances at volume: the numbers that size any migration

400<sub>k</sub>

TOP-LEVEL INSTANCES IN 90 DAYS; 1.28M INCLUDING CALL-ACTIVITY CHILDREN

7,943

APPLICATIONS ON THE BUSIEST DAY (23,865 ENGINE INSTANCES); 1,347 IN THE BUSIEST HOUR

1.10<sub>M</sub>

ENGINE INSTANCES OPEN; 1.04M OPEN USER TASKS; 46M RUNTIME VARIABLES

490<sub>GB</sub>

OF ENGINE HISTORY AT FULL LEVEL, 90-DAY RETENTION, ABOUT 1.4M ACTIVITY ROWS A DAY

| PROCESS KEY, 90 DAYS          | STARTED |  | OPEN    | MEAN  |
|-------------------------------|---------|--|---------|-------|
| NDF2W                         | 253,728 |  | 100,633 | 54 h  |
| NDF4W                         | 61,930  |  | 17,298  | 99 h  |
| OPERATION                     | 21,999  |  | 2,395   | 25 h  |
| Unified_Process_Main_Workflow | 20,092  |  | 9,079   | 142 h |
| NDF4W_RO                      | 11,713  |  | 2,592   | 43 h  |

THERE IS NO DRAIN WINDOW

540,824 top-level applications are open. 408,638 of them started more than 90 days ago and 59,710 more than a year ago. One NDF2W survey wait state alone holds 695,734 open tasks. **Any plan must migrate in-flight state or run two engines side by side while the old book runs off.** 4,433 open incidents are retried by six people from Cockpit.

# Camunda 7 Enterprise: what we think it is, and what we need confirmed

Our understanding: swap Community artifacts for Enterprise, move 7.23 to 7.24, add a licence, gain a patch channel and support. Models, delegates, database and consoles stay as they are. The schema log already reads 7.23.0 cleanly.

|   |                                                                                                                                                                                                                     |                                            |
|---|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|
| 1 | <b>Support horizon.</b> The published end date for 7.24 Enterprise patches, and what happens after it.                                                                                                              | DECIDES WHETHER A IS A<br>BRIDGE OR A HOME |
| 2 | <b>Spring Boot.</b> Which Boot lines the 7.24 Enterprise starter supports, and whether a Spring Boot 4 starter is on the Camunda 7 roadmap. Boot 3.5 is out of open-source support; this is our binding constraint. | THE KEY QUESTION                           |
| 3 | <b>Upgrade path.</b> Database steps beyond the schema-log row; behaviour changes to test against our escalation, link and retry patterns.                                                                           | EFFORT INPUT                               |
| 4 | <b>Our extensions.</b> A custom <code>ReadOnlyIdentityProvider</code> plugin and seven Cockpit plugin bundles. Anything in Enterprise Cockpit that affects them.                                                    | COMPATIBILITY                              |
| 5 | <b>Features we would use.</b> Batch operations on 4,433 incidents, instance migration tooling, history views, and whatever reduces 490 GB of history on Cloud SQL.                                                  | VALUE                                      |
| 6 | <b>Licensing scope and effort.</b> Two production deployments, four environments each, plus the unrelated 7.17 install. Your effort estimate for A, assuming we do the Spring Boot work.                            | COMMERCIAL, SEPARATE<br>THREAD             |

# Camunda 8 Enterprise: a re-platforming, and the eight things we most need assessed

|   |                                                                                                                                                                                                                                                                                                                         |                               |
|---|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| 1 | <b>Model conversion.</b> Run the Migration Analyzer on our 53 models. What the Diagram Converter handles and what needs a hand: 659 JUEL conditions, 239 mappings, 190 escalation elements, 194 link events, 4 inclusive joins, 56 propagate-all call activities, 186 retry cycles, 3 DMN tables with FIRST hit policy. | WE SUPPLY MODELS UNDER NDA    |
| 2 | <b>Delegates to workers.</b> 272 classes, 483 bindings, all Spring beans. Adapter, Spring SDK workers, or generated worker per delegate? What the adapter does not cover: <code>BpmnError</code> from 88 sites, <code>hasVariable</code> , modification from 5 places.                                                  | LARGEST WORK ITEM             |
| 3 | <b>Transactions.</b> The shared-transaction model on slide 6 becomes idempotent workers. Recommended pattern, and how much of it is generic.                                                                                                                                                                            | LARGEST RISK ITEM             |
| 4 | <b>In-flight state.</b> 541k open applications, 1.1M engine instances, 1.0M open user tasks, tail to a year. What the Data Migrator covers for runtime and user tasks from 7.23 or 7.24, or whether the plan is two engines and route new applications.                                                                 | DECIDES THE SHAPE OF THE PLAN |
| 5 | <b>History and sizing.</b> 1.4M activity rows a day, 90-day floor. Storage sizing for 8.8; whether history migrates or we accept a cut-over point. SaaS with an Indonesian data-residency answer, or Self-Managed on GKE.                                                                                               | HOSTING                       |
| 6 | <b>Human tasks and consoles.</b> We need wait state, assignment by variable, and search/claim/complete through an API. Which surface do we build against so three consoles keep working?                                                                                                                                | NO TASKLIST, NO FORMS         |
| 7 | <b>Operations.</b> Keycloak OIDC for Operate, role mapping, what replaces our seven Cockpit plugins, incident retry at scale, the equivalent of process instance modification.                                                                                                                                          | OPS STAFF                     |
| 8 | <b>Enablement and effort.</b> Phased estimate, the services or partner model you recommend, training for squads who know 7 well and 8 not at all.                                                                                                                                                                       | PEOPLE                        |

# What does not move, whichever path we take

|                                                                                                                                                                                                                                                                    |                                                                                                                                                                      |                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div><div>REGULATED WORKLOAD</div><div><b>Auditability and retention are floors</b></div><div>Loan origination records for a supervised lender. 90 days of process history is the minimum we keep, and decisions must stay explainable after the fact.</div></div> |                                                                                                                                                                      | <div><div>NO DRAIN WINDOW</div><div><b>In-flight state is the plan, not a footnote</b></div><div>Half a million open applications with a one-year tail. Every deployment today is already a hot deployment against live engine state.</div></div> | <div><div>HUMAN WORK IS OURS</div><div><b>Three consoles, zero Camunda references</b></div><div>The operator experience is built and will not be rebuilt on Tasklist. The engine must expose the wait state and an API. That is all the UI tier asks of it.</div></div>                                                                                                                                                                                           |
| <div><div>PLATFORM</div><div><b>Google Cloud, Indonesia</b></div><div>GKE, Cloud SQL PostgreSQL 16, Keycloak. Data residency matters for any SaaS answer.</div></div>                                                                                              | <div><div>TWO DEPLOYMENTS</div><div><b>Conventional and Sharia</b></div><div>Same code, separate databases and pipelines, separate regulatory treatment.</div></div> | <div><div>TEAM</div><div><b>Java and Spring squads</b></div><div>Deep Camunda 7 habits, no Camunda 8 or Zeebe experience, thin automated coverage at the process layer.</div></div>                                                               | <div><div>WHAT WE CAN SHARE, AND WHAT WE CANNOT</div><div><b>Can:</b> all 53 BPMN and 3 DMN files under NDA, anonymised database statistics beyond the ones in this package, and a read-only walk-through of the code with an engineer.</div><div><b>Cannot:</b> production data, credentials, hostnames, or customer records. <b>Not in this package by choice:</b> internal cost figures and vendor comparisons. Those are a separate conversation.</div></div> |

# Four weeks from package to written assessment

|                                                                                                                                                                                            |                                                                                                                                                                                                              |                                                                                                                                                                                  |                                                                                                                                                                                                   |
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| <b>WEEK 1</b><br><b>Walk-through</b><br>One call. We take you through this package and the five files, agree the model hand-over and the NDA, and name one technical contact on each side. | <b>WEEKS 1-2</b><br><b>Analysis</b><br>You run the Migration Analyzer on our models and read the integration and footprint files. We answer questions within a day and give a code walk-through if it helps. | <b>WEEKS 3-4</b><br><b>Written assessment</b><br>One document per path: what changes, effort ranges, the risks you see, and the questions on slides 8 and 9 answered in writing. | <b>WEEK 4</b><br><b>Review</b><br>A second call with our engineers and the people who operate the system. Then we decide, and the commercial conversation starts from a shared technical picture. |
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|          |                                                                                                                                                                        |                |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| <b>A</b> | <b>Reference customers</b> at similar scale, 100,000-plus long-running instances a month, one for each path.                                                           | HELPFUL        |
| <b>B</b> | <b>Coexistence.</b> If we take Path A now and Path B later, what a Camunda 7 Enterprise subscription contributes to the Camunda 8 move, contractually and technically. | DECISION INPUT |
| <b>C</b> | <b>Support horizon and Spring Boot answer first.</b> Slide 8, items 1 and 2, can be answered before the analysis. They shape how we read everything else.              | WEEK 1         |

**HOW WE WILL BOTH KNOW IT WORKED**  
We can put the two paths side by side with effort, risk and timeline from the same evidence. The migration plan, on either path, starts from element counts and production numbers both sides have already agreed rather than from a discovery phase. And the first workshop is about our system, not about a slide template.